LINUX REMOTE LOGIN AND CLI

LINUX

- What is Linux?
  - Started out as a hobby project built from scratch using C by Linus Torvalds
  - Developed by the community to be similar to UNIX

- What about UNIX?
  - Originally an operating system
  - Modified(forked) into different variants by people
  - Currently both a trademark and a specification of the Open Group consortium
  - An OS that is certified by the Open Group may be called 'UNIX'
    - e.g. Apple OS-X, Oracle Solaris, IBM AIX, HP-UX
  - Linux is merely 'UNIX-like' → It's not certified

LINUX

- Linux as an **open Source** software
  - Means that people are allowed to obtain the code and modify it to suit their needs
  - The code is not owned by any company
  - It is developed by the community that uses it

- Components of Linux as an operating system
  - Kernel
    - Central controller that manages hardware resources for the use of applications
  - Applications
    - Software that make requests to the kernel for resources to perform tasks
    - *Process* is a task that is loaded and tracked by the kernel
    - Applications may run 1 or more processes

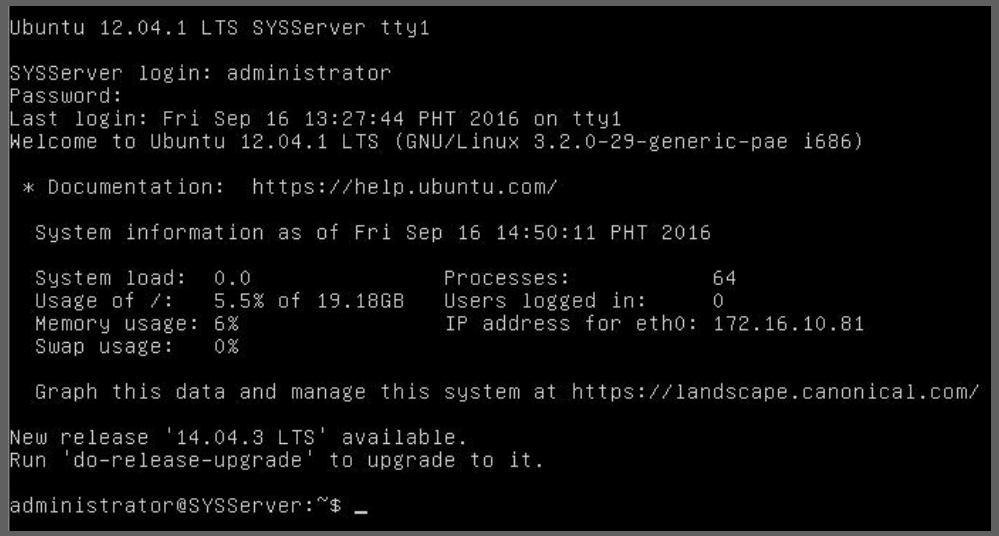
LINUX DISTRIBUTIONS

- Are operating systems that share the same Linux kernel, but differ by the set of tools , applications, and package managers that they come with
- May come readily complied for installation on specific architectures or may be in code form
- Some common distros
  - Red Hat – built to be more suited for use on servers, offers paid support
  - Open SUSE – targetted for software developers and system administrators
  - Debian – community backed distribution which directly supports several non-Intel / AMD platforms
  - Ubuntu – most popular desktop / server distro derived from Debian
  - Android – uses the Linux kernel, but lacks support for many tools

USING LINUX

- Graphical Mode

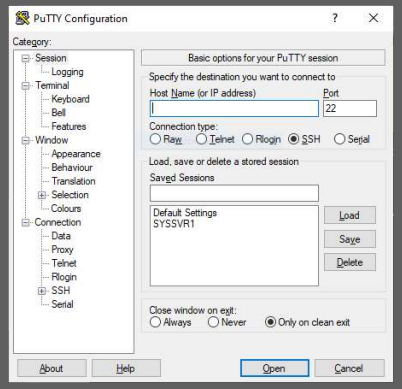
- Non- Graphical Mode

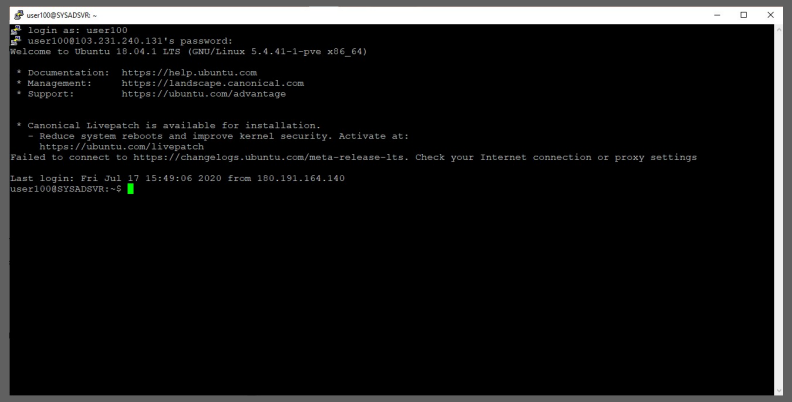


```
Ubuntu 12.04.1 LTS SYSServer tty1

SYSServer login: administrator
Password:
Last login: Fri Sep 16 13:27:44 PHT 2016 on tty1
Welcome to Ubuntu 12.04.1 LTS (GNU/Linux 3.2.0-29-generic-pae i686)

 * Documentation:  https://help.ubuntu.com/

  System information as of Fri Sep 16 14:50:11 PHT 2016

  System load:  0.0               Processes:           64
  Usage of /:   5.5% of 19.18GB   Users logged in:     0
  Memory usage: 6%                IP address for eth0: 172.16.10.81
  Swap usage:   0%

  Graph this data and manage this system at https://landscape.canonical.com/

New release '14.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

administrator@SYSServer:~$ _
```

REMOTE LOGIN

- Remote login to Unix and Linux is through the SSH protocol
  - SSH Protocol – Secure Shell Protocol
- In Windows, the PuTTY application can be used to connect remotely
  - PuTTY can be downloaded at https://www.putty.org/
- To login, a user account is needed.
  - A user account has a username and password





THE LINUX COMMAND LINE

- Why is learning the command line important?
  - More precise control
  - Greater speed
  - Ability to automate tasks

- What is a 'shell'?
  - Interpreter that translates commands entered by the user into actions performed by the OS
  - There are many types of shells; one of the most common is the BASH shell

THE LINUX COMMAND LINE

- What is a 'prompt'?
  - Terminal window displays a prompt when no commands are being run
  - Tells a user to enter a command

```
administrator@SYSServer:~$
```

Logged in user

System name

Current directory

(~ means home dir)

THE LINUX COMMAND LINE

- What is a 'command'?
  - It's a software program that when executed, causes the OS to run a process

- Typical format: **`command [options] [arguments]`**
  - Options modify core command behavior
    - Single letter preceded by a dash e.g. "ls -l -h" or "ls -lh"
    - Full-word preceded by double dash e.g. "ls --human-readable"
  - Arguments provide additional information

KEY SEQUENCES

Key Sequence	Description
Ctrl-c	Terminates or aborts a process
Ctrl-d	Sends an end-of-input on the keyboard
Ctrl-s	Pauses screen output
Ctrl-q	Continues screen output
Tab	Auto-complete files and folder names
Up / Down arrows	Cycle through command history

CHANGING PASSWORDS

- To change password, the `passwd` command is used
- After typing in the command, the system will ask for the current password
- Then will ask for the new password twice
  - Second time is for verification

INTRODUCTION TO LINUX AND THE COMMAND LINE INTERFACE